

(2 Hours)

[Total Marks: 60]

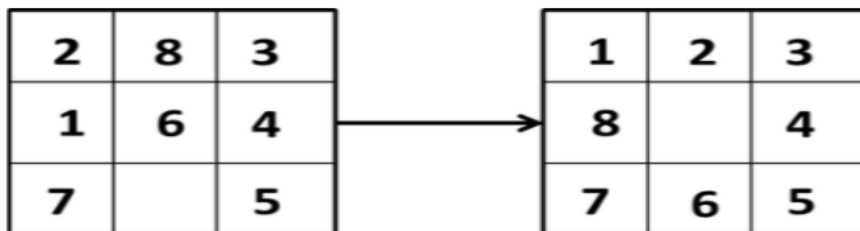
- N. B.: (1) All questions are compulsory.
 (2) Make suitable assumptions wherever necessary and state the assumptions made.
 (3) Answers to the same question must be written together.
 (4) Numbers to the right indicate marks.
 (5) Draw neat labeled diagrams wherever necessary.
 (6) Use of Non-programmable calculators is allowed.

1. **Attempt any two of the following:** 12

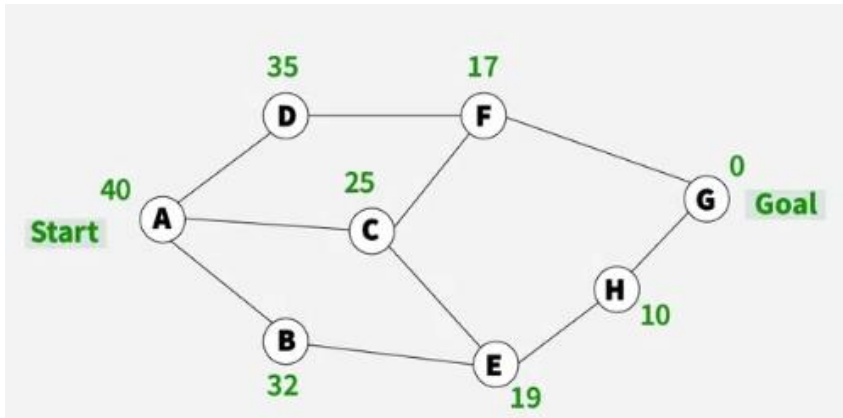
- a. Discuss the historical evolution of Artificial Intelligence.
- b. Describe the Turing test, designed to provide an operational definition of Artificial Intelligence.
- c. Define the following terms with examples:-
 - i) Agent
 - ii) Rational agent
 - iii) Irrational agent
 - iv) Learning agent
 - v) Omniscience agent
 - vi) Autonomy agent
- d. Explain the utility-based agent with a neat labeled diagram.

2. **Attempt any two of the following:** 12

- a. Differentiate between Uninformed and Informed Search Strategies.
- b. Formulate 8-puzzle problem for the following scenario and also mention the steps and path cost required to reach goal state.

**Initial State****Goal State**

- c. Provide an overview of the Greedy Best-First Search algorithm. Using the following graph, where nodes represent states and edges represent connections with heuristic values, find the optimal path from node A (start state) to node G (goal state) using the Greedy Best-First Search algorithm.

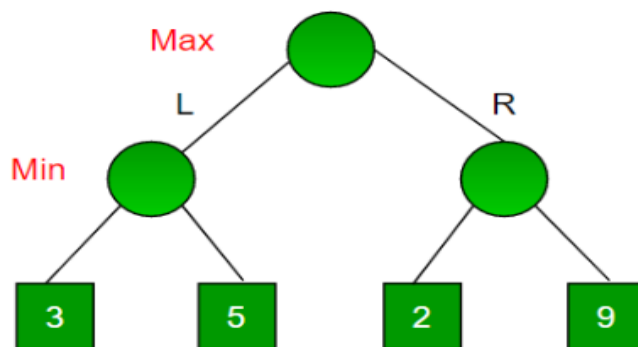


- d. Explain the State-space search diagram for the Hill Climbing algorithm in detail.

3. **Attempt any two of the following:**

12

- Explain various techniques required to get the best optimal solution and also mention various elements of Game Playing search.
- Explain the minimax algorithm, highlighting its advantages, disadvantages, and properties. Also, evaluate the given search tree and determine the max values at root level.



- Explain and differentiate between two types of Inference systems for Knowledge-Based Agents (KBA).
- Explain PEAS descriptors for WUMPUS world and also mention its different properties.

4. **Attempt any two of the following:**

12

- Define First Order Logic and explain its syntax and semantics.
- Describe in detail knowledge engineering process using in First Order Logic.
- Convert the below given Facts into FOL and Prove that "Colonel is a Criminal" using **Forward-Chaining**.

- It is crime for an American to sell weapons to the enemy of America.
- Country Nono is an enemy of America.
- Nono has some Missiles.
- All the Missiles were sold to Nono by Colonel.
- Missile is a weapon.
- Colonel is American.

- d. Define Resolution and also summarize the steps for Resolution. Consider the statements:-
- Rimi is hungry.
 - If Rimi is hungry she barks.
 - If Rimi is barking then Raja is angry.

Explain statements in predicate logic. Convert them into CNF form. Prove that "**Raja is angry**?" using Resolution. And also draw Resolution tree.

5. **Attempt any two of the following:**

- a. What is classical planning in AI? Provide an explanation with an example. Also, **12**
describe the goal of planning using a supermarket scenario.
- b. Explain progression and regression state-space search algorithms.
- c. Write short notes on Categories and Objects, and Mental Events and Mental Objects.
- d. Describe the different languages used for the implementation of planning in brief.
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